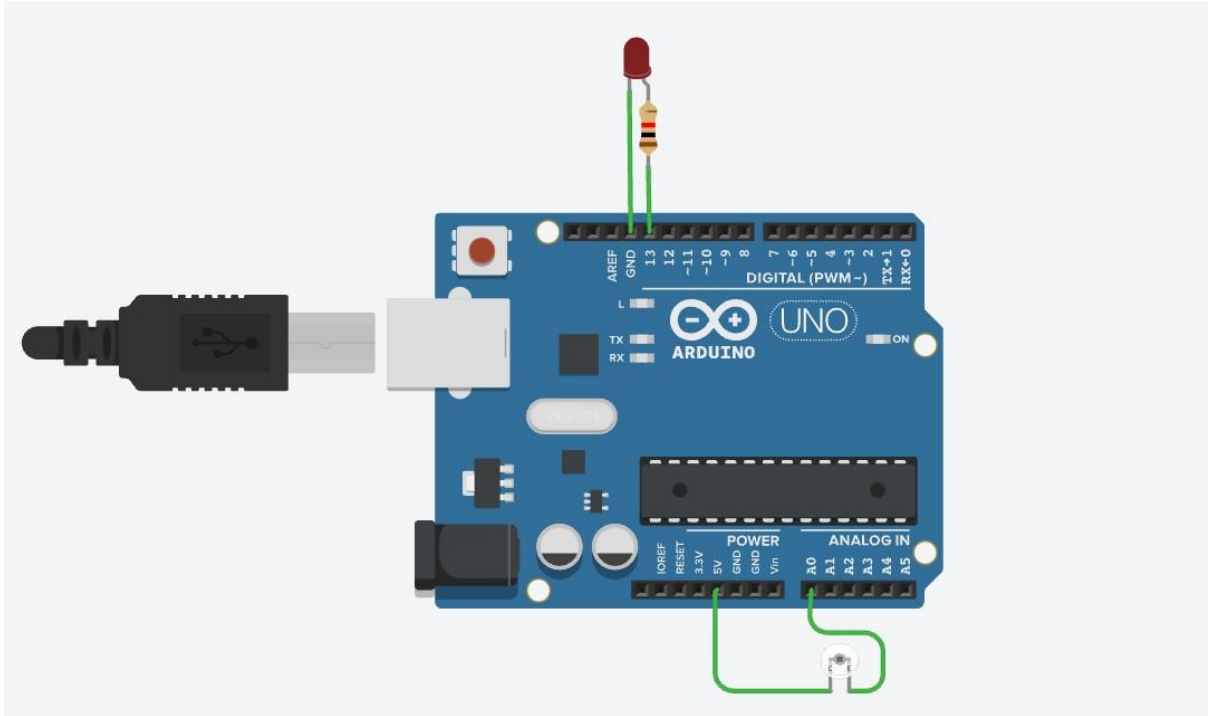


Embedded System and Internet of Things

Task:-03

1. Circuit Diagram:-



2. Source Code:-

```
int photoPin = A0;    // Phototransistor output

int ledPin = 13;      // LED pin
int threshold = 500;  // Light threshold value

void setup() {
    pinMode(ledPin, OUTPUT);
}
```

```
Serial.begin(9600);

}

void loop() {  int lightValue = analogRead(photoPin); //
Read sensor value

Serial.print("Light Value: ");

Serial.println(lightValue);

if (lightValue > threshold) {  // Bright condition
digitalWrite(ledPin, HIGH);  // LED ON
Serial.println("Bright Light - LED ON");
} else {  // Dark condition
digitalWrite(ledPin, LOW);  // LED OFF
Serial.println("Low Light - LED OFF");
}

delay(500);

}
```

3. Explanation:-

A) How the Phototransistor Works:-

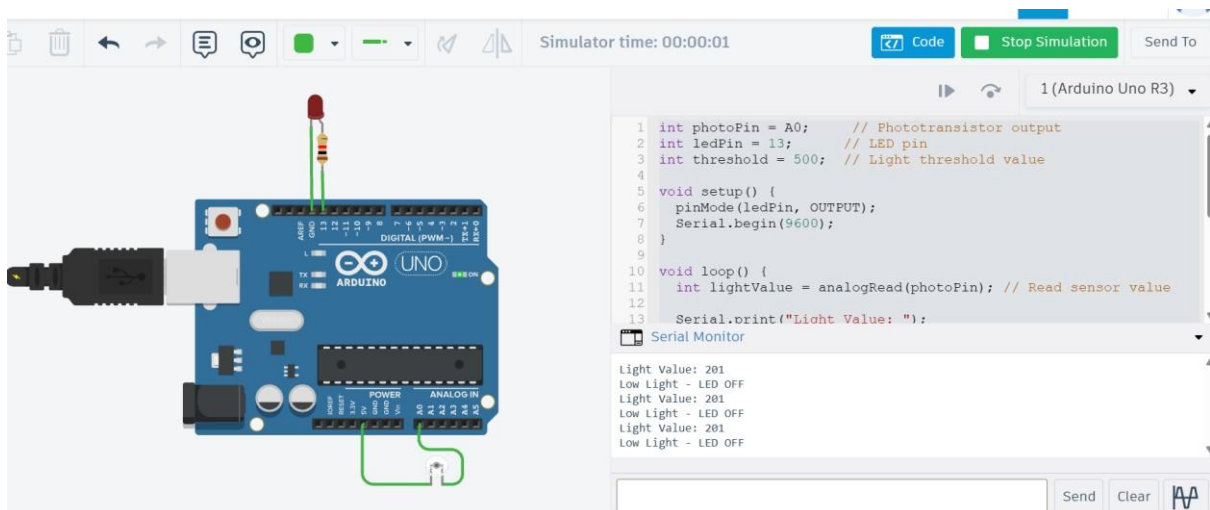
- A phototransistor changes its **current flow based on light intensity**.
- More light → more current → higher voltage at Arduino analog pin.
- Less light → lower voltage.

B) What the Code Does:-

- Reads light intensity using `analogRead()`
- Displays the value on **Serial Monitor**
- Compares it with a **threshold**
- Turns **LED ON** when light exceeds threshold
- Turns **LED OFF** otherwise

4. Output:- 1)

Led OFF:-



2) Led ON:-

